

Wind power forecasting based on variational mode decomposition and high-order fuzzy cognitive maps

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ABSTRACT

Accurate wind power forecasting can effectively reduce the adverse effects of wind power forecasting errors on wind power grid integration and power dispatch. However, current wind power forecasting technology, such as the method based on machine learning, belongs to the black box model and is not solvable. Variational mode decomposition (VMD) is a decomposition technique based on the time-frequency characteristics of the original time series, which has a mathematical theoretical foundation. Besides, fuzzy cognitive map (FCMs) is a kind of soft computing method with strong knowledge representation and reasoning ability. Therefore, to enhance the forecasting accuracy of wind power, in this paper, a novel time series forecasting method based on improved VMD (IVMD) and high-order FCM (HFCM), namely IVMD_{HFCM} is proposed. IVMD can effectively extract the features in the raw time series depending on the time-frequency characteristics of the time series. Then, the subseries obtained by IVMD are modeled and forecasted by HFCM, and the Bayesian ridge regression method is adopted to learn the weight of HFCM. Finally, the differential evolution (DE) algorithm is used to get the optimal hyperparameters of IVMD_{HFCM}. The performance of IVMD_{HFCM} is verified by comparing it with that of state-of-the-art methods on ten publicly available datasets. Moreover, the proposed IVMD_{HFCM} is compared with the existing HFCM based method on ten actual wind power datasets. The results show that the IVMD_{HFCM} can effectively improve the accuracy of wind power forecasting and reduce the forecasting error. Besides, the IVMD_{HFCM} can also effectively explore the fluctuations of wind power.

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1. Introduction

Energy is an essential basis for the survival and development of human society. With the increasing exhaustion of traditional fossil energy such as oil and coal, as well as the deterioration of the environment and the increasing electricity demand, a new energy revolution is quietly emerging [1]. However, the key to the energy revolution is to vigorously develop renewable energy such as wind power and solar energy. In fact, wind power is growing very rapidly all over the world. Nowadays, the global installed wind power capacity was approximately 743 GW, an increase of 14.3% over 2019. In 2020, the global new installed wind power capacity was 93 GW, an increase of 53% year-on-year [2]. In China, the accumulative capacity of wind power was 281 GW, with 71.67 GW of new capacity installed in 2020 [3]. However, the nature of randomness, volatility, and intermittency of wind power may limit the capacity incorporation into the power grid

and the absorption ratio. One of the key measures to solve this problem is to promote accurate wind power forecasting methods.

In recent decades, many successful wind power forecasting methods have been developed to improve forecasting accuracy, which can be classified into four categories, namely physical methods, statistical methods, machine learning methods, and hybrid methods [4,5]. Generally, the physical methods were used in long-term forecasting, which is based on numerical weather prediction (NWP). Therefore, a large number of physical factors, such as terrain description, surface roughness, atmospheric density, pressure, and temperature, are needed to achieve optimal forecasting accuracy [6]. Statistical methods are mainly time series prediction models, such as Kalman filters [7], autoregressive (AR) [8], autoregressive moving average (ARMA) [9], autoregressive integrated moving average (ARIMA) [10], generalized autoregressive conditional heteroskedasticity (GARCH) [11], and their extended versions. These methods are used to forecast the future output of wind power based on the vast number of historical data, which are more effective than physical methods for short-term wind power forecasting. With the rapid development of artificial intelligence in recent years, many prediction techniques based on machine learning [12], such as recurrent neural networks

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(RNN) [13], artificial neural networks (ANN) [14], deep neural networks (DNN) [15], long short-term memory (LSTM) [16], and gated recurrent unit (GRU) [17], have been developed to deal with wind speed or wind power forecasting. However, these methods require the training of a vast number of historical data to improve forecasting accuracy.

To continually improve the forecasting accuracy, the hybrid method combining the advantages of different prediction models was proposed. Xia et al. in [18] proposed the fusion of the neural network and fuzzy logic for wind power forecasting. Wang et al. [19] improved forecasting accuracy by integrating wavelet transform (WT) and echo state network (ESN). Catalao et al. proposed a variety of hybrid forecasting methods for short-term wind power forecasting in Portugal based on WT, particle swarm optimization (PSO), neural network, and fuzzy logic [20–22]. Unlike WT, Zhang et al. [23] used the empirical mode decomposition (EMD) method to decompose the wind power data into several intrinsic mode functions (IMF) and a residual component, then trained each component with a support vector machine (SVM), and finally accumulate each component to get the final predicted value. In recent years, researchers have developed different hybrid prediction models based on EMD, such as the hybrid methods based on EMD and chaos theory [24], based on EMD and relevance vector machine (RVM) [25], based on EMD and radial basis function neural network (RBFNN) [26], based on EMD and bagging neural network [27], and based on EMD and deep convolutional neural network (CNN) [28]. Because EMD lacks a mathematical foundation, the variational mode decomposition (VMD) technology was developed to replace EMD [29]. Many hybrid methods based on VMD have been successfully developed [30–32]. As we all know, machine learning is a black-box model, which requires certain prior knowledge and training on a large number of historical data to achieve a certain prediction accuracy.

Fuzzy cognitive maps (FCMs) are a kind of soft computing method with stronger knowledge representation and reasoning ability [33]. There have been many successful applications of FCM and high-order FCM (HFCM) in time series forecasting [34–37]. In order to significantly reduce the forecasting error of wind power and make the forecasting model interpretable, in this paper, a new model based on improved VMD and HFCM (IVMD_{HFCM}) is developed to forecast wind power. In order to build a complete FCM, the VMD is improved by adding a residual term, referring to the EMD. To the best of our knowledge, this is the first research that uses FCM for wind power forecasting and the first time that VMD and HFCM have been combined for time series forecasting.

In the proposed IVMD_{HFCM} method, the original time series is decomposed by IVMD to obtain $K+1$ subseries (including K IMFs and one residual term), which are utilized to construct h -order HFCMs with $K+1$ nodes. Once the structure of the HFCM is constructed, it can be used to predict future values. The detailed framework of the proposed IVMD_{HFCM} is presented in Section 3.4. The number of IMFs K and penalty factor α are two important parameters of IVMD. If the value of K is too small, insufficient decomposition may occur. α directly affects the accuracy of the decomposition. The higher the value of α is, the wider the bandwidth of K IMFs will be. On the contrary, the smaller the value of α is, the smaller the bandwidth of K IMFs will be, which will affect the decomposition accuracy. Besides, the orders h in HFCM also has a certain influence on forecasting accuracy. However, most methods based on HFCM use grid search method to find the optimal parameters [34–37], which is very time-consuming. For parameters in VMD, some studies use evolutionary computing methods to get corresponding parameters according to different fitness functions [38,39]. However, these fitness functions do not use the proposed IVMD_{HFCM} method. Therefore, in this paper,

the differential evolution (DE) algorithm is used to find the optimal hyperparameter pair for the proposed IVMD_{HFCM}, and the minimum root-mean-square error (RMSE) is used as the fitness function.

The proposed IVMD_{HFCM} can be divided into three stages to deal with time series forecasting. Firstly, HFCM is constructed by subseries obtained from the raw time series decomposed by IVMD; Secondly, the weight matrix of HFCM is learned to predict future values. In this paper, the Bayesian ridge regression is used to learn the weight matrix; Finally, RMSE is calculated and DE is used to find the optimal hyperparameter pair. To verify the performance of IVMD_{HFCM}, which is first compared with state-of-art methods on ten public datasets. The results show that IVMD_{HFCM} can significantly improve the forecasting accuracy. Then, IVMD_{HFCM} is compared with the existing HFCM based method on ten different actual wind power datasets to further study its performance. The results demonstrate that IVMD_{HFCM} can sense the abrupt data points in the time series so as to track the fluctuation of wind power. Consequently, IVMD_{HFCM} has the ability to improve the forecasting of wind power with volatility, randomness, and intermittence, thereby reducing the error of wind power forecasting.

The rest of this paper is organized as follows. Section 2 briefly introduces the related work on VMD and FCMs. The proposed IVMD_{HFCM} is detailed in Section 3. Section 4 gives the experiment and discussion. Finally, the conclusion is given in Section 5.

2. Related work

This section gives a brief overview of the VMD and FCMs used in this paper.

2.1. Variational mode decomposition

VMD is a completely non-recursive signal decomposition algorithm [29], which can decompose the original series/signal $f(t)$ into K subseries or u_k ($k = 1, 2, \dots, K$) modes. Each mode u_k is compact around a center frequency ω_k , which is determined by the decomposition. A complete decomposition should be a minimum of the sum of the frequency bandwidths of each mode while satisfying that the sum of each mode u_k is equal to the original series/signal $f(t)$. For $f(t)$, the corresponding constrained variational problem model is given as follows:

$$\min_{\{u_k\}, \{\omega_k\}} \left\{ \sum_{k=1}^K \left\| \delta_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \quad (1)$$

$$s.t. \sum_{k=1}^K u_k = f(t)$$

where $\{u_k\} := \{u_1, u_2, \dots, u_K\}$ and $\{\omega_k\} := \{\omega_1, \omega_2, \dots, \omega_K\}$ are the set of all modes and center frequencies, respectively. $\delta(t)$ is the Dirac distribution, and δ_t is the partial derivative concerning time t . The notation of $*$ indicates convolution. j is the imaginary unit.

To solve the above variational problem with constraint, the quadratic penalty factor α and Lagrangian multiplier λ are used to convert it into an unconstrained one. α can ensure the accuracy of the series reconstruction, especially in the presence of Gaussian noise. The augmented Lagrangian function is given as follows,

$$L(\{u_k\}, \{\omega_k\}, \lambda) = \alpha \sum_{k=1}^K \left\| \delta_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 + \left\| f(t) - \sum_{k=1}^K u_k \right\|_2^2 + \left\langle \lambda(t), f(t) - \sum_{k=1}^K u_k \right\rangle \quad (2)$$

To find the saddle point of (2) as the solution to the original variational problem, the alternate direction method of multipliers (ADMM) is used to solve the above Lagrangian function. The saddle point of (2) is obtained by iteratively updating $un+1$ k , ω_{n+1} k , and λ^{n+1} based on ADMM, n is the number of iterations. The final updated formulas for $\hat{u}_k^{n+1}$ and ω_{n+1} k are expressed as (3) and (4), respectively.

$$\hat{u}_k^{n+1}(\omega) = \frac{\hat{f}(\omega) - \sum_{i \neq k} \hat{u}_i(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k)^2} \quad (3)$$

$$\omega_k^{n+1} = \frac{\int_0^\infty \omega |\hat{u}_k(\omega)|^2 d\omega}{\int_0^\infty |\hat{u}_k(\omega)|^2 d\omega} \quad (4)$$

where $\hat{\cdot}$ denote the Fourier transform. Finally, $un+1$ k is obtained by inverse the Fourier transform of the real part of (3).

For all $\omega \geq 0$, using the dual ascent method to update $\hat{\lambda}^{n+1}$, and the formula is given as follows,

$$\hat{\lambda}^{n+1}(\omega) = \hat{\lambda}^n(\omega) + \tau \left(\hat{f}(\omega) - \sum_{k=1}^K \hat{u}_k^{n+1}(\omega) \right) \quad (5)$$

where τ is the update factor, generally, it is set to 0. Similarly, λ^{n+1} can be obtained by the inverse Fourier transform.

When (6) is satisfied or the maximum number of iterations (N_{max_ADMM}) is reached, ADMM terminates the iteration. ε is the preset tolerance.

$$\sum_{k=1}^K \frac{\|\hat{u}_k^{n+1} - \hat{u}_k^n\|_2^2}{\|\hat{u}_k^n\|_2^2} < \varepsilon \quad (6)$$

The process of decomposing a time series into K subseries by VMD is shown in Algorithm 1.

Algorithm 1: VMD

Input:

$f(t)$: The original series;
 K : The number of IMFs;
 α : The penalty factor;
 ε : The tolerance;
 N_{max_ADMM} : The number of iterations;

Output:

$\{u_1, u_2, \dots, u_K\}$;

Initialize $\{u_k^1\}, \{\omega_k^1\}, \hat{\lambda}^1, n \leftarrow 1$;

while $n \leq N_{max_ADMM}$ **or** (6) is unsatisfied **do**

for $k=1$ **to** K **do**

 Update $\hat{u}_k(\omega)$ using (3);

 Update ω_k using (4);

end for;

 Update $\hat{\lambda}$ according to (5);

$n \leftarrow n+1$;

end while;

2.2. Fuzzy cognitive maps

The topology of FCMs is denoted by a directed weighted graph whose nodes and edges represent the concepts and fuzzy logic relationships among concepts, respectively. The concept state value vector $\mathbf{A}$ of an FCM with N_c concepts or nodes is defined as follows,

$$\mathbf{A} = [A_1, A_2, \dots, A_{N_c}] \quad (7)$$

where the state value $A_i \in [0, 1]$ or $[-1, 1]$ ($i = 1, 2, \dots, N_c$) represents the activation degree of node i . The fuzzy logic relationships

among concepts are denoted by a matrix $\mathbf{W}$ with size $N_c \times N_c$.

$$\mathbf{W} = \begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1N_c} \\ w_{21} & w_{22} & \cdots & w_{2N_c} \\ \vdots & \vdots & \ddots & \vdots \\ w_{N_c1} & w_{N_c2} & \cdots & w_{N_cN_c} \end{bmatrix} \quad (8)$$

where $w_{ij} \in [-1, 1]$ ($i, j = 1, 2, \dots, N_c$) represents the causal influence exerted by node j upon node i . $w_{ij} = 0$ means that there is no causal relationship between nodes i and j . $w_{ij} > 0$ indicates that the increase of node i will lead to the increase of node j ; that is, there is a positive causal relationship between nodes i and j . Conversely, $w_{ij} < 0$ indicates a negative causal relationship.

In FCMs, the state value of node i at time t can be denoted as $A_i(t)$. The state value at time $t+1$ is determined by the weighted matrix and the state values of all nodes at time t . Therefore, the dynamics of FCMs can be defined as follows,

$$A_i(t+1) = g \left(\sum_{j=1}^{N_c} w_{ij} A_j(t) \right) \quad (9)$$

where $g(\bullet)$ is a transfer function to map the activation level into unit range. The sigmoid function and the hyperbolic tangent are the two most widely used transfer functions in FCMs [40]. The former can map the output to $[0, 1]$, and the hyperbolic tangent can map the output state values to the range of $[-1, 1]$. In this paper, since the VMD is employed to decompose the original time series, only the hyperbolic tangent can be used, which is defined as follows,

$$g(x) = \frac{e^x + e^{-x}}{e^x - e^{-x}} \quad (10)$$

According to (9), the state value of nodes at time $t+1$ only depends on the state values of all nodes at time t ; that is, FCMs only model short-term temporal relationships, ignoring the influence of the earlier state values on the current state value. To compensate for the drawback, Stach et al. in [41] developed the HFCMs to model the long temporal dependency of nodes. An h -orders HFCMs can be defined as follows [36],

$$A_i(t+1) = g \left(\sum_{j=1}^{N_c} w_{ij}^1 A_j(t) + w_{ij}^2 A_j(t-1) + \cdots + w_{ij}^h A_j(t-h+1) + w_{i0} \right) \quad (11)$$

where w_{ij}^s ($s = 1, 2, \dots, h$) represents the causal influence exerted by node j upon node i at time step s . w_{i0} is the constant bias term related to the i th node, whose range lies in $[0, 1]$.

3. Proposed method

3.1. Improved variational mode decomposition

In this paper, the subseries obtained from VMD decomposition is used as the HFCM nodes. Therefore, the decomposition precision of the original series directly affects the accuracy of state value $A_i(t+1)$ solved by HFCM. Ideally, in the conventional VMD method, K IMFs obtained by ADMM can be reconstructed into the original series according to (1). Because the condition for ADMM to stop iterating is meeting N_{max_ADMM} or (6), this may make the decomposition incomplete when the iteration is terminated. Besides, the conventional VMD has the problem of edge effect and the existing researches always require the high symmetry

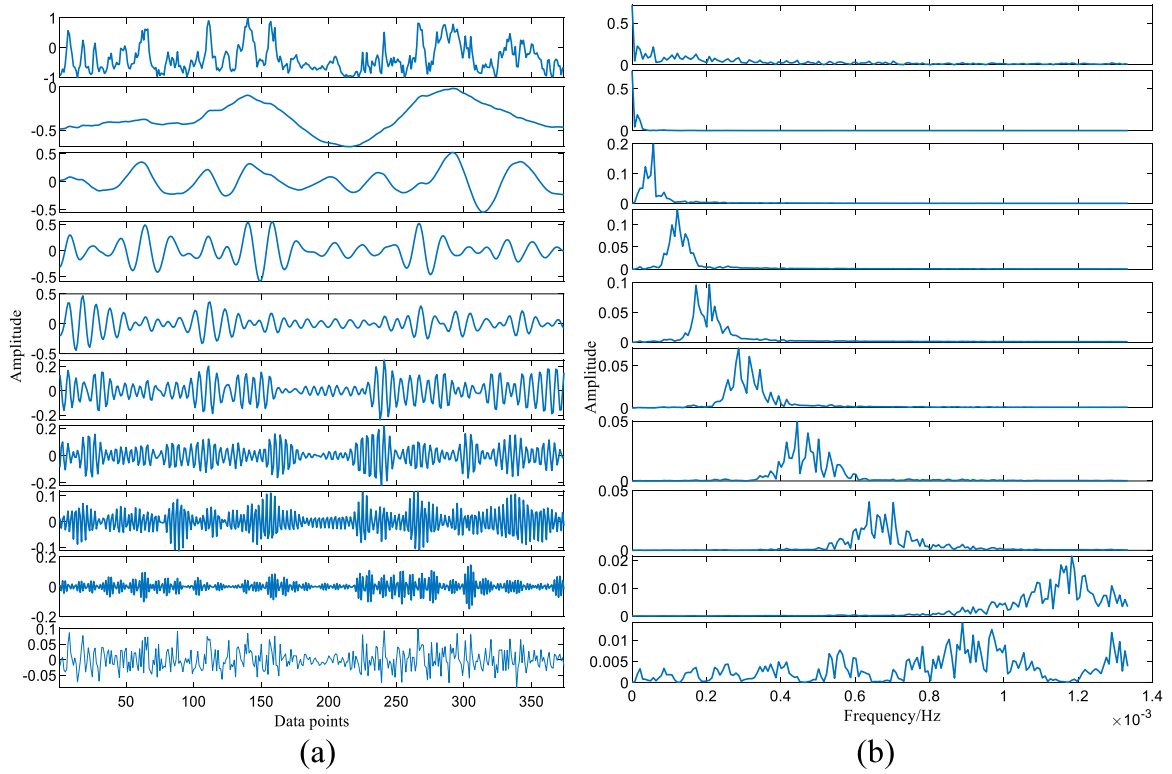


Fig. 1. The decomposition results of the wind power dataset by IVMD. In (a), from top to bottom are the original dataset, IMF1, IMF2, IMF3, IMF4, IMF5, IMF6, IMF7, IMF8, and Res, respectively. (b) is the spectrum corresponding to each sequence in (a).

of the original input data [42]. In fact, the data collected in engineering applications are usually irregular and asymmetrical (e.g., the time series of wind power has strong intermittence and fluctuation). This also directly affects the reconstruction accuracy after decomposition. Moreover, this incomplete decomposition may produce serious cumulative errors, thereby reducing the accuracy of state value $A_i(t+1)$. Based on the above analysis, in order to eliminate the incompleteness of VMD decomposition, we improved the VMD (IVMD); that is, a residual (Res) term is introduced after VMD decomposition. Both [41] and Res are taken as nodes of HFCM, which improve the accuracy of state value $A_i(t+1)$. The Res term is defined as follows,

$$Res(t) = f(t) - \sum_{i=1}^K u_k \quad (12)$$

where $Res(t)$ is the residual after conducting the VMD decomposition at time t . To visualize the IVMD decomposition effect, one wind power dataset as an example is decomposed by IVMD. For the sake of simplicity, the parameters of K and α are set to 8 and 467, respectively. Therefore, eight IMFs and one Res are obtained by IVMD, which are nine subseries shown in Fig. 1(a). Besides, Fig. 1(b) shows the corresponding spectrum of each subseries. It is clear from Fig. 1 that the IVMD decomposes the original dataset according to its different frequencies.

3.2. HFCM learning from time series

In this paper, the subseries of the original series decomposed by IVMD are used as the nodes of HFCM. Then the key of the proposed forecasting method is used to learn the weight of HFCM. The learning problem of the weight matrix of FCM proposed in [43] was simplified to the local connection learning of every single node, which is the most popular HFCM weight learning framework. The same weight learning framework is employed

in this paper, which is shown in Fig. 2. Researchers have proposed many effective weighing learning methods, such as the Hebbian learning method [44–46], evolutionary computing methods [43,47,48], and methods based on regression [36,37,49]. In these methods, Hebbian methods are mainly suitable for domain experts to determine the initial weight and then refine the weight. Evolutionary computing is a population-based optimization method, which is stochastic and time-consuming, so it is difficult to use for large-scale time series. Regression-based methods such as the lasso regression [49], ridge regression [36], and Bayesian ridge regression [37], can effectively learn the weight only by tuning parameters. However, compared with lasso regression and ridge regression, the Bayesian ridge regression has been proved to be less sensitive to the parameters. Consequently, Bayesian ridge regression is used to learn the weight matrix of HFCM in this paper.

The form of the dynamic transformation process of HFCM state values is shown in (11), which is equivalent to (13).

$$g^{-1}(A_i(t+1)) = \sum_{j=1}^{N_c} (w_{ij}^1 A_j(t) + w_{ij}^2 A_j(t-1) + \dots + w_{ij}^h A_j(t-h+1) + w_{i0}) \quad (13)$$

where $g^{-1}()$ is the inverse function. Let vector $\mathbf{Y}_i$ denote $g^{-1}(A_i(t+1))$ at different t . The vector $\mathbf{X}$ denotes the state values $A_j(t)$ of all nodes. The vector $\mathbf{W}_i$ represents the weight vectors of all nodes to nodes i . Therefore, (13) can be converted to $\mathbf{Y}_i = \mathbf{XW}_i$ vector form. For a time series of length L , in HFCM with N_c nodes, $\mathbf{Y}_i$, $\mathbf{X}$, and $\mathbf{W}_i$ are expressed as given in Eqs. (14)–(16) in Box I.

The weight matrix of HFCM can be obtained by the Bayesian ridge regression method. The detail of the Bayesian ridge regression model was shown in [37,50], and the objective function is defined as follows,

$$\min_{\mathbf{W}_i} = \left\{ \frac{\beta}{2L} \|\mathbf{Y}_i - \mathbf{XW}_i\|_2^2 + \frac{\gamma}{2} \|\mathbf{W}_i\|_2^2 \right\} \quad (17)$$

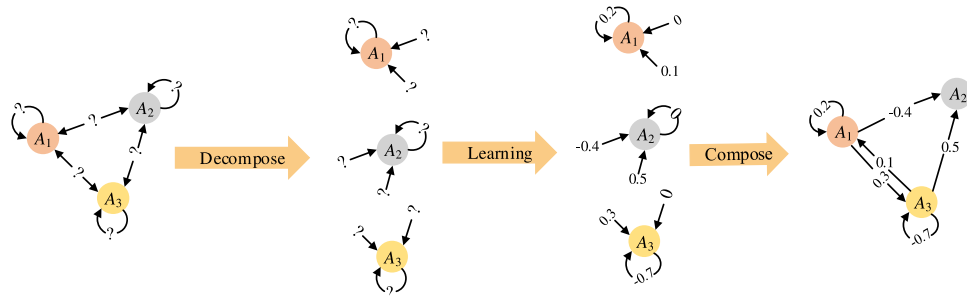


Fig. 2. The framework of FCM weight matrix learning.

$$\mathbf{Y}_i = [g^{-1}(A_i(h)), g^{-1}(A_i(h+1)), \dots, g^{-1}(A_i(L))]^T. \quad (14)$$

$$\mathbf{X} = \begin{bmatrix} A_1(h) & \dots & A_1(1) & A_2(h) & \dots & A_2(1) & \dots & A_{N_c}(h) & \dots & A_{N_c}(1) & 1 \\ A_1(h+1) & \dots & A_1(2) & A_2(h+1) & \dots & A_2(2) & \dots & A_{N_c}(h+1) & \dots & A_{N_c}(2) & 1 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots \\ A_1(L-1) & \dots & A_1(L-h) & A_2(L-1) & \dots & A_2(L-h) & \dots & A_{N_c}(L-1) & \dots & A_{N_c}(L-h) & 1 \end{bmatrix} \quad (15)$$

$$\mathbf{W}_i = [w_{i1}^1 \quad \dots \quad w_{i1}^h \quad w_{i2}^1 \quad \dots \quad w_{i2}^h \quad \dots \quad w_{iN_c}^1 \quad \dots \quad w_{iN_c}^h \quad w_{i0}]^T \quad (16)$$

Box I.

where β and γ are the regularization parameters, which are suggested to set to 10^{-6} in [51].

3.3. Hyperparameters optimization of IVMD_{HFCM} via DE algorithm

The VMD is not adaptive, and the modulus K and penalty parameter α are necessary to set in advance. Besides, large parameter deviations may lead to loss of modes or mixing of signal components, which limits its universality. The existing research methods of adjusting parameters are not applicable in this paper. Order h is the most critical parameter of HFCM, and appropriate h can make the weight matrix learning more accurate, thus reducing the prediction error. Therefore, K , α , and h are the hyperparameters that need to be optimized in IVMD_{HFCM}. The grid search method is very time-consuming in tuning three hyperparameters. DE is a heuristic random search algorithm based on population individual differences [52]. It is a simple but powerful optimizer, insensitive to the convexity-concavity and continuity of the functions, and is often used for solving parameter tuning problems and engineering optimization problems [43,44]. Consequently, the relatively efficient DE algorithm is used to find the optimal hyperparameters in this paper. The RMSE of the validation set obtained by IVMD_{HFCM} is defined as the objective function of the DE algorithm, which is expressed as follows,

$$RMSE = \sqrt{\frac{\sum_{i=1}^L (x_i - \hat{x}_i)^2}{L}} \quad (18)$$

where x_i and $\hat{x}_i$ are the raw data and prediction value, respectively. L is the length of the dataset. The process (K, α, h) is shown in Algorithm 2.

3.4. The framework of IVMD_{HFCM}

In the IVMD_{HFCM}, the raw dataset is divided into three subsets; that is, training set, validation set, and test set. The training set is

used to learn the weight matrix from the HFCM. The validation set is used to choose the best model, and the test set is used to evaluate the performance of the selected model. Besides, to avoid some unnecessary numerical problems that may affect the forecasting accuracy of IVMD_{HFCM}, it is necessary to normalize the original series. VMD is a decomposition method based on the time-frequency technology, therefore, the original series needs to be mapped to $[-1, 1]$. Consequently, the max-min normalization method is employed in this paper, which is defined as follows,

$$\bar{x} = 2 \times \frac{x - x_{\min}}{x_{\max} - x_{\min}} - 1 \quad (19)$$

where $x_{\max}$ and $x_{\min}$ are the maximum and minimum values of the corresponding time series x . Finally, the forecasting values need to be renormalized to properly evaluate the performance of IVMD_{HFCM}. The overall framework of IVMD_{HFCM} is shown in Fig. 3. The optimal hyperparameters are obtained by the DE algorithm, and the IVMD_{HFCM} with these hyperparameters is the best forecasting model. Finally, the forecasting value obtained by IVMD_{HFCM} should be renormalized to evaluate the performance by calculating the RMSE.

4. Experiments and discussion

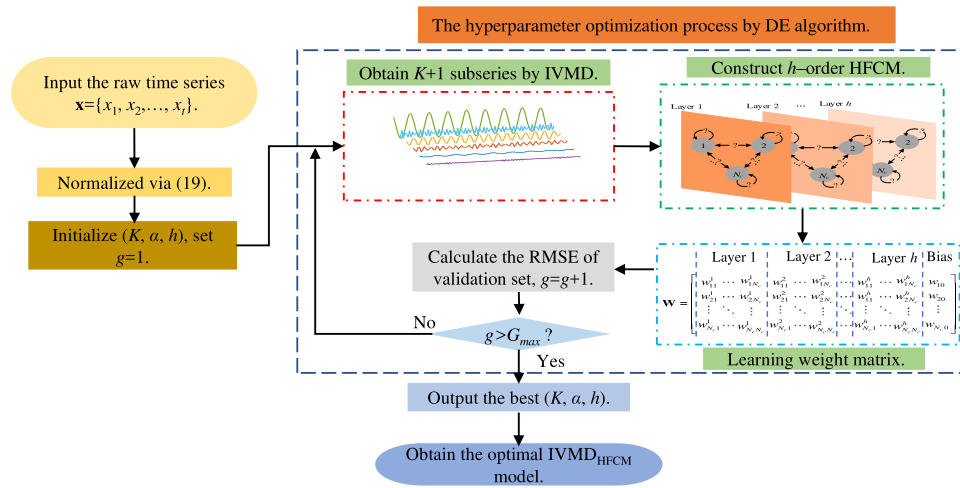
In this section, the performance of IVMD_{HFCM} is first verified by comparing it with that of state-of-art methods on ten public datasets. Then, the proposed IVMD_{HFCM} is applied to ten different actual wind power time series data to further study its performance. Finally, the sensitivity of the parameters in the IVMD_{HFCM} is analyzed.

4.1. Datasets

(1) *Public datasets*: Ten frequently used public datasets with different characteristics are used to verify IVMD_{HFCM}'s performance. The first one is the Wolf sunspot data, which is the

Algorithm 2: Hyperparameter optimization of (K, α, h) by DE**Input:**The train data set S_t , and the validation set S_v ;**Output:** (K, α, h) .

- 1 Initialize K, α , and h ;
- 2 For each pair (K, α) , K subseries of S_t is obtained by Algorithm 1;
- 3 Calculate the Res term by (12);
- 4 Construct FCM nodes $A_j(t), j=1, 2, \dots, K+1$ (K subseries and one Res);
- 5 Construct h -orders HFCM, then obtain vectors $\mathbf{Y}_{S_t}$ and $\mathbf{X}_{S_t}$ by (14) and (15);
- 6 Learn the weight matrix $\mathbf{W}$ by the Bayesian ridge regression method;
- 7 For S_v , perform lines 2-5, and generate the h -orders HFCM of S_v , then obtained $\mathbf{X}_{S_v}$;
- 8 $\mathbf{Y}_{S_v} \leftarrow \mathbf{X}_{S_v} \mathbf{W}$;
- 9 Summing $\mathbf{Y}_{S_v}$, get the predicted value $\hat{S}_v$ of S_v ;
- 10 Calculate RMSE by (18);
- 11 Update K, α , and h by DE algorithm;
- 12 For the new K, α , and h , execute lines 2-10 until the terminal criterion of DE is met;

**Fig. 3.** The overall framework of the proposed IVMD_{HFCM} method.

annual number of sunspots observed from 1700 to 1988, and its dataset length is 288 [53]. The second is the daily opening price of the S&P500 index from Jun. 1, 2016 to Jun. 1, 2017, which contains 251 observed data [54]. The third dataset recorded the monthly milk production (MMP) from Jan. 1962 to Dec. 1975, which contains 168 data [55]. The fourth dataset is the closing monthly Dow Jones industrial index (DJII) from Aug. 1968 to Aug. 1981, with a total of 291 data points [56]. The fifth dataset collected the monthly average concentration of CO₂ (ppm) in Mauna Loa (CML) from 1965 to 1980, the data length is 192 [57]. The sixth dataset length is 536, which records the daily Istanbul stock exchange national-100 (ISEN100) index from Jun. 5, 2009 to Feb. 22, 2011 [58]. The seventh dataset contains 600 data, which record the monthly Lake Erie (MLE) from 1921 to 1970 [59]. The eighth dataset contains 1034 data points, which are hourly room temperature (RTemp) data collected by a monitoring system [37]. The ninth dataset is the monthly critical radio frequencies (MCRF) in Washington, D.C. from May 1934 to April 1954, which contains 240 pieces of data [60]. The last dataset is the chaotic Mackey–Glass (MG) time series [61], which is a common benchmark for testing the performance of forecasting models. In this paper, the MG dataset contains 1000 data from $t = 124$ to $t = 1123$ is generated based on (20) by the 4-order Runge Kutta algorithm.

Generally, $x(0)$ and τ are set to 1.2 and 17 [36,37], respectively.

$$\dot{x}(t) = \frac{0.2x(t - \tau)}{1 + x^{10}(t - \tau)} - 0.1x(t) \quad (20)$$

The characteristics of the minimum (Min), maximum (Max), mean and standard deviation features of each dataset are shown in Table 1. Besides, the size of the subsets obtained by dividing each dataset is also listed in Table 1.

(2) *Wind power datasets*: Wind power datasets were collected from different wind farms with different sampling periods. The first wind farm data to be studied comes from a public dataset, which collected wind power for the whole of 2011 with a resolution of 10 min [62]. In this wind farm, three days in each quarter are selected to form datasets to study ultra-short-term wind power forecasts. These data were collected from 2/12/2011–2/14/2011, 5/7/2011–5/9/2011, 8/8/2011–8/10/2011, and 12/10/2011–12/12/2011, and each dataset contains 432 data. The four datasets in ultra-short-term forecasting are abbreviated as USA, USB, USC, and USD, respectively. The data of the second wind farm is derived from the measured data of wind farm A in Northwest China in 2016 (sampled every 1 h), which is divided into four different datasets according to the quarter to study short-term wind power forecasting. These datasets are named as SA, SB, SC, and SD. The rated installed capacity of wind farm A is 90MW. To study the medium-short-term forecasting of wind

Table 1

The feature of public datasets and the size of subsets.

Datasets	Min	Max	Mean	Std	Training(%)	Validation(%)	Test(%)
Sunspot	0	190.2	48.61	39.47	177(61.46)	44(15.28)	67(23.26)
S&P500	1833.4	2270.5	2094.3	101.23	120(48)	30(12)	101(40)
MMP	553	969	754.71	102.21	108(64)	26(16)	34(20)
DJII	3573	3978	3754.86	105.47	175(60)	43(15)	73(25)
CML	317.25	341.19	328.46	5.96	131(68)	32(17)	29(15)
ISEN100	-0.0622	0.069	0.0016	0.0163	343(64)	85(16)	108(20)
MLE	10	20	14.99	2.01	368(61.4)	92(15.3)	140(23.3)
RTemp	11.39	28.91	20.46	3.12	620(60)	155(15)	259(25)
MCRF	4.3	13.5	8.08	2.44	176(73.6)	44(18.4)	20(8)
MG	0.2192	1.3137	0.9194	0.236	400(40)	100(10)	500(50)

Table 2

Length and features of wind power time series.

Datasets	Length	Min	Max	Mean	Std	Training(64%)	Validation(16%)	Test(20%)
USA(kW)	373	0.48	405.26	130.91	91.96	239	59	75
USB(kW)	359	1.91	393.30	142.92	82.60	230	57	72
USC(kW)	374	0.48	394.26	129.85	91.18	240	59	75
USD(kW)	311	4.31	396.65	181.96	100.87	199	49	63
SA(MW)	2183	0.08	98.80	10.56	14.13	1397	349	437
SB(MW)	2179	0.01	69.50	14.47	13.83	1395	348	436
SC(MW)	2099	0.10	69.20	11.71	10.77	1344	335	420
SD(MW)	2204	0.04	102.00	6.39	11.13	1411	352	441
MSA(MW)	1244	0.10	81.00	13.40	11.96	795	199	250
MSB(MW)	1196	0.15	62.80	13.01	10.97	765	191	240

power, the daily average generating capacity of wind farms A and B in Northwest China from 2014 to 2017 were collected. These two datasets are abbreviated as MSA and MSB.

The actual wind power data recorded will produce outliers due to sensor failure or shutdown of the generator. Therefore, outliers in the dataset need to be eliminated before the prediction starts. All the time series of wind power after removing outliers and the characteristics of each dataset are shown in Table 2. Besides, the sizes of training, validation, and test datasets are 64%, 16%, and 20%.

4.2. Experimental setup

In IVMD, parameters N_{max_ADMM} and ε are used as those suggested in [29], which are 500 and $1e-7$, respectively. For the DE parameters, the scaling factor F and crossover rate CR are set to 0.9 and 0.3 [63], respectively. G_{max} is set to 100, and N_{pop} is 15. For the sake of simplicity, the parameters range of K , α , and h are set to [2, 20], [1, 5000], and [1, 20], respectively.

The RMSE is used as the performance criteria to evaluate the forecasting performance of IVMD_{HFCM}. Besides, P_{RMSE} is used to evaluate the forecasting performance of any two methods, which is defined as follows,

$$P_{RMSE} = \frac{RMSE_{com} - RMSE_{our}}{RMSE_{com}} \times 100\% \quad (21)$$

where $RMSE_{our}$ and $RMSE_{com}$ are RMSE values obtained by IVMD_{HFCM} and the corresponding comparison method, respectively. The results of P_{RMSE} denote that the percentage improvement in the performance of IVMD_{HFCM} compared to the comparison method.

All datasets are normalized to $[-1, 1]$ using (19) before starting to optimize the hyperparameters. Besides, all experiments are performed in the Python environment on a PC with Core I7-10700 CPU(2.90 GHz), and 24 G RAM. In each dataset, the DE algorithm performs 11 independent runs and the best hyperparameters are recorded. Note that RMSE in all experiments involving algorithm comparisons is calculated based on the test set.

4.3. Experiments on public datasets

To verify the performance of IVMD_{HFCM}, some state-of-art methods include sparse autoencoder (SA) FCM [34], EMD-HFCM [37], Kernel-HFCM [35], Wavelet-HFCM [36], adaptive-network-based fuzzy inference system (ANFIS) [64], AR model [65], Multiresolution AR (MAR) model [66] and ANN [67], are used in the comparison on the same subset level. The values of RMSE obtained by all methods are listed in Table 3, in which the bold represents the best value. Besides, since the computing time of the comparison methods does not give in the corresponding literature, only the computing time of the proposed IVMD_{HFCM} is listed in Table 3. Note that the calculation time after fixing K and α parameters is given in the brackets. From Table 3 we can see, except for the MG dataset, the RMSE of IVMD_{HFCM} is lower than that of other methods in nine datasets. In terms of the MG dataset, the RMSE obtained by IVMD_{HFCM} is consistent with the ANFIS, only slightly worse than that of SAE-FCM. Besides, the RMSE of IVMD_{HFCM} on the MG dataset is also the lowest compared with that of other methods except for SAE-FCM and ANFIS. Consequently, it is reasonable to conclude that IVMD_{HFCM} can effectively improve the accuracy of time series forecasting. Besides, the computing cost is also within the acceptable range.

The P_{RMSE} values of the other methods compared to that of IVMD_{HFCM} are given in Table 4. It can be seen that the performance of the IVMD_{HFCM} is significantly improved compared with other methods. Especially, in the sunspots, ISEN100, MLE, and MCRF datasets, the performance improvement of the IVMD_{HFCM} compared with the existing method has exceeded 90%. This further shows that the performance of the IVMD_{HFCM} is superior. Besides, compared with those based on the HFCM method, namely, SAE-FCM, EMD-HFCM, Kernel-HFCM, and Wavelet-HFCM, the performance of the proposed IVMD-based HFCM method is significantly improved on all datasets except for slightly inferior to SAE-FCM on the MG dataset. This shows that IVMD, a method based on data frequency decomposition, can fully mine the inherent features of data compared to other data feature extraction techniques, thereby improving the performance of HFCM in time series forecasting.

The hyperparameters of each dataset and the RMSE of the corresponding training, validation, and test subsets are given in

Table 3The computing time (s) of the IVMD_{HFCM} and the values of RMSE by all methods.

Datasets	IVMD _{HFCM}	SAE-FCM	EMD-HFCM	Kernel-HFCM	Wavelet-HFCM	ANFIS	AR	MAR	ANN
Sunspot	2.490 (3035(90))	17.390	17.216	21.236	18.916	22.753	35.262	19.186	19.901
S&P500	6.122 (2973(27))	11.893	7.139	13.701	16.105	14.935	17.897	16.041	17.696
MMP	3.004 (1822(61))	7.931	7.403	12.321	8.258	9.578	57.717	37.838	27.113
DJII	4.765 (4939(31))	21.335	18.875	23.411	23.159	27.526	29.822	26.733	28.52
CML	0.151 (7614(71))	0.366	0.923	0.549	0.560	0.910	0.902	0.662	0.652
ISEN100	0.00002 (13543(164))	0.013	0.011	0.013	0.015	0.014	0.018	0.015	0.015
MLE	0.034 (5138(150))	0.359	0.436	0.385	0.377	0.458	0.638	0.390	0.402
RTemp	0.081 (19755(209))	0.165	0.116	0.146	0.224	0.157	0.579	0.124	0.421
MCRF	0.034 (6680(90))	0.490	0.989	0.611	0.547	0.651	0.902	0.662	0.652
MG	0.001 (3578(46))	0.0009	0.009	0.002	0.004	0.001	0.035	0.002	0.005

Table 4 P_{RMSE} (%) of IVMD_{HFCM} compared with that of other methods.

Datasets	SAE-FCM	EMD-HFCM	Kernel-HFCM	Wavelet-HFCM	ANFIS	AR	MAR	ANN
Sunspot	85.68	85.54	88.27	86.84	89.06	92.94	87.02	87.49
S&P500	48.52	14.25	55.32	61.99	59.01	665.79	61.84	65.40
MMP	62.12	59.42	75.62	63.62	68.64	94.80	92.06	88.92
DJII	77.67	74.75	79.65	79.42	82.69	84.02	82.18	83.29
CML	58.74	83.64	72.50	73.04	83.41	83.26	77.19	76.84
ISEN100	99.85	99.82	99.85	99.84	99.86	99.89	99.87	99.87
MLE	90.53	92.20	91.17	90.98	92.58	94.67	91.28	91.54
RTemp	50.91	30.17	44.52	63.84	48.41	86.01	34.68	80.76
MCRF	93.06	96.56	94.44	93.78	94.78	96.23	94.86	94.79
MG	-11.11	88.89	50.00	75.00	0	97.14	50.00	80.00

Table 5RMSE of IVMD_{HFCM} on each subset and the corresponding hyperparameters for each dataset.

Datasets	Training	Validation	Test	(K, α , h)
Sunspot	1.050	0.645	2.490	(18, 7, 10)
S&P500	2.668	6.690	6.122	(10, 1500, 6)
MMP	0.406	1.923	3.004	(17, 21.5, 18)
DJII	5.522	6.312	4.765	(9, 98, 10)
CML	0.204	0.061	0.151	(18, 60.5, 13)
ISEN100	0.00002	0.00003	0.00002	(20, 3.5, 20)
MLE	0.025	0.018	0.034	(19, 44, 11)
RTemp	0.041	0.055	0.081	(20, 82, 2)
MCRF	0.056	0.056	0.034	(19, 690, 5)
MG	0.0007	0.0009	0.001	(10, 2878, 5)

Table 5. Fig. 4 shows the original dataset and predicted time series obtained by IVMD_{HFCM}. It can be seen that the predicted values obtained by IVMD_{HFCM} can fully learn the variation trend of the original dataset. Therefore, by comparing the state-of-art methods on ten datasets, it can be concluded that IVMD_{HFCM} in this paper is effective in time series forecasting and can significantly improve the accuracy of prediction.

Besides, the evaluation indicators of the worst, median, best, average, and standard deviation (Std) are added to analyze the performance of the IVMD_{HFCM}. The RMSE and hyperparameters corresponding to the worst, median, best, average, and standard deviation (Std) are listed in Table 6. Compared with the results in Tables 3 and 6, the values of Std indicator obtained on all datasets are small, which indicates that IVMD_{HFCM} is robust when using DE to tune hyperparameters. On public datasets, except the MG dataset, the RMSE obtained by IVMD_{HFCM} is the best in both the worst, median, best, and average indicators compared with other methods. For the MG dataset, in all indicators, IVMD_{HFCM} is only slightly worse than SAE-FCM. In the worst indicator, IVMD_{HFCM} is better than AR and achieves a similar result as EMD-HFCM. In the median indicator, IVMD_{HFCM} is worse than SAE-HFCM and ANFIS and gets the same effect as Kernel-HFCM and MAR. Besides, in the average indicators, IVMD_{HFCM} is the best compared with EMD-HFCM, Wavelet-HFCM, AR, and ANN. Consequently, the proposed IVMD_{HFCM} forecasting method has high prediction accuracy on public datasets.

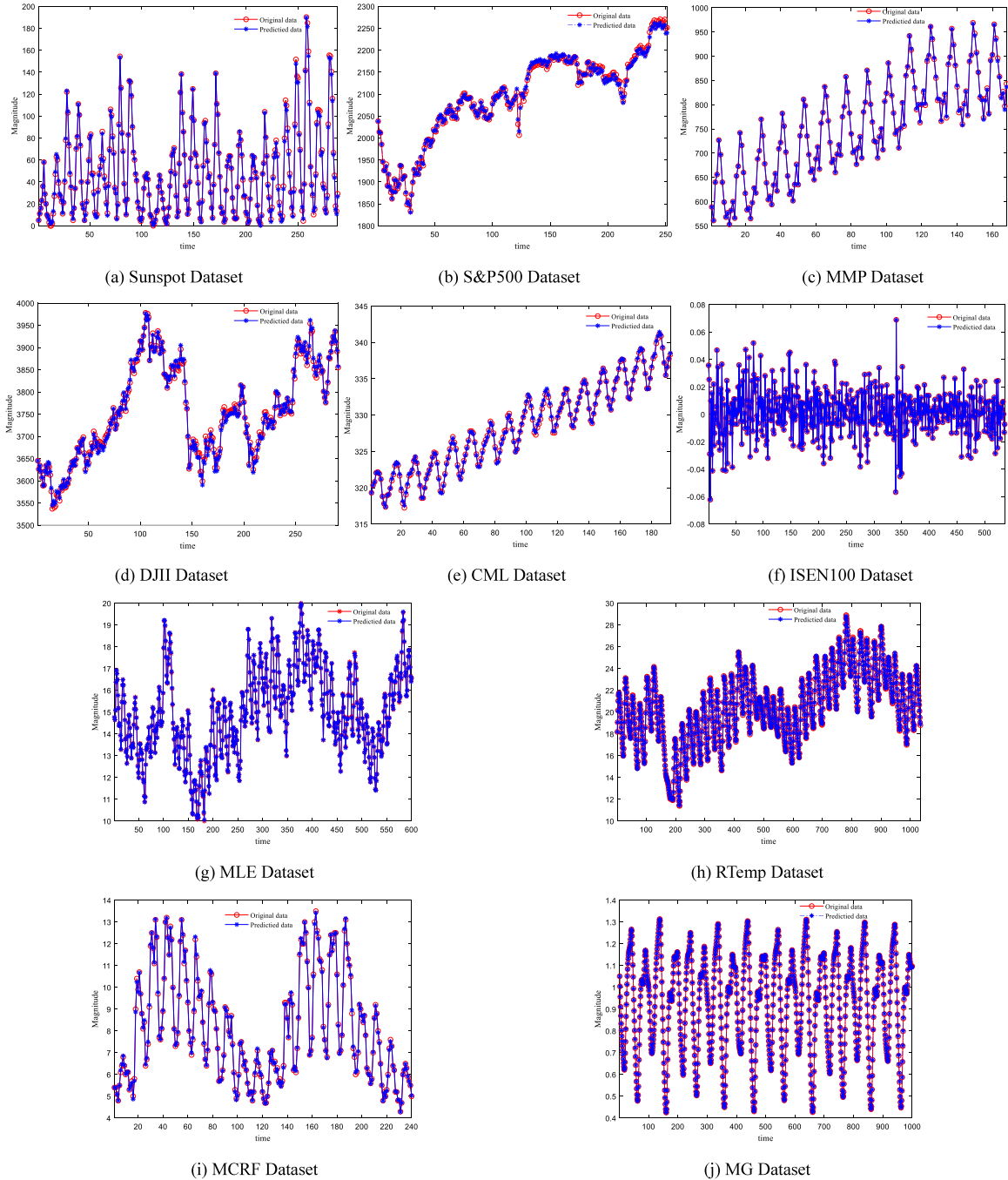
4.4. Wind power forecasting

In this section, three different types of wind power forecasting, namely, ultra-short-term, short-term, and medium-short-term forecasting of wind power, are studied in ten wind power datasets actually collected. These datasets are split into different types by different sampling frequencies. IVMD_{HFCM} is used to forecast the different datasets according to the dataset divided in Table 2. Besides, to demonstrate the forecasting performance of IVMD_{HFCM}, some latest HFCM-based time series forecasting methods, namely SAE-FCM, EMD-HFCM, Kernel-HFCM, and Wavelet-HFCM methods, are used to predict the wind power datasets under the same case. To make the comparison credible, these comparison methods use the grid search method to record the optimal results, and the parameters involved come from the corresponding literature. The RMSE and computing time of all methods in wind power datasets are listed in Table 7. From Table 7, the RMSE obtained by IVMD_{HFCM} is the lowest in all wind power datasets. Therefore, IVMD_{HFCM} is excellently effective in wind power forecasting.

For computing time, the EMD-HFCM and Wavelet-HFCM have fewer hyperparameters, so the computing time is shorter. Besides, in EMD-HFCM, the IMF of the dataset obtained by the EMD method needs to meet certain conditions, which limits the number of subsets obtained by the original dataset decomposition. In EMD-HFCM, the EMD toolkit is used, and the number of decompositions per dataset is fixed by the default parameters

Table 6The evaluation indicators of the IVMD_{HFCM} on public datasets.

Datasets	Worst(K, α, h)	Median(K, α, h)	Best(K, α, h)	Average	Std
Sunspot	2.707(20,37,20)	2.645(19,36,20)	2.490(18, 7, 10)	2.615	0.066
S&P500	7.976(13,17,20)	7.338(12,58,20)	6.122(10, 1500, 6)	7.194	0.578
MMP	6.380(17,7,20)	3.465(18,23,13)	3.004(17, 21.5, 18)	3.755	0.903
DJII	5.970(7,51,5,6)	5.568(9,38,5,9)	4.765(9, 98, 10)	5.532	0.354
CML	0.364(10,16,5,20)	0.348(9,11,20)	0.151(18, 60.5, 13)	0.300	0.084
ISEN100	0.0000251(20,4,19)	0.0000245(20,4.5,18)	0.0000223(20, 3.5, 20)	0.0000239	0.000001
MLE	0.051(20,367,15)	0.049(20,49.5,20)	0.034(19, 44, 11)	0.046	0.006
RTemp	0.109(20,1872,14)	0.086(20,106,20)	0.081(20, 82, 2)	0.095	0.013
MCRF	0.141(15,12,8)	0.0348(19,1703,5)	0.034(19, 690, 5)	0.063	0.043
MG	0.009(17,899.5,16)	0.002(12,2892.5,6)	0.001(10, 2878, 5)	0.003	0.002

**Fig. 4.** Original time series and predicted time series obtained by IVMD_{HFCM}.

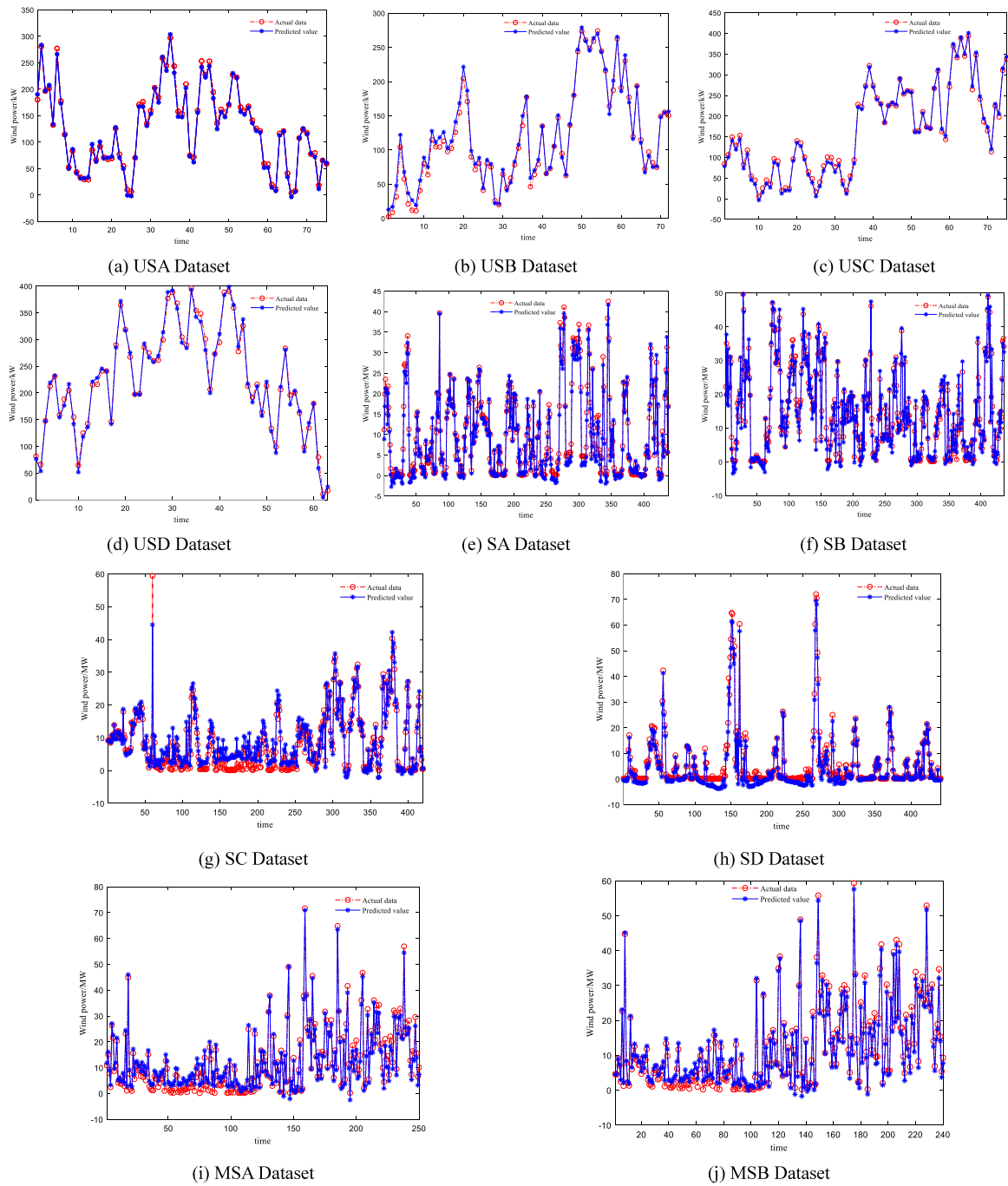


Fig. 5. Actual data and predicted value of wind power obtained by $IVMD_{HFCM}$. USA, USB, USC, and USD datasets are ultra-short-term wind power forecasting datasets; SA, SB, SC, and SD datasets are short-term wind power forecasting datasets; MA and MB datasets represent medium-short-term forecasting datasets.

of the toolkit; that is, the number of nodes in HFCM is fixed. Therefore, the computing of the EMD-HFCM is the least. In the proposed $IVMD_{HFCM}$, the DE is used to find the best hyperparameters. Therefore, compared with EMD-HFCM and Wavelet-HFCM, it is time-consuming in the model training phase. In addition, we also give the computing time of the $IVMD_{HFCM}$ when the VMD parameters (obtained by DE optimization) are fixed, which are listed in the parentheses after the total computing time in Table 7. The reason for the high computing time, in this case, is that the increase in the number of subsets derived from decomposition also increases the computing cost. Since there are many hyperparameters in SAE-HFCM and Kernel-HFCM, and the grid search method is used, the computing cost is more expensive than other methods. Although the computing time of the proposed

$IVMD_{HFCM}$ is not the best compared to other methods. However, the proposed $IVMD_{HFCM}$ method has the best prediction accuracy within an acceptable time frame.

The improving percentage P_{RMSE} of the $IVMD_{HFCM}$ method compared with other methods is given in Table 8. In Table 2, the standard deviation of the ultra-short-term forecasting datasets is relatively high, indicating that the data is unstable. However, on these datasets, the RMSE improvement of $IVMD_{HFCM}$ is more than 70% compared with other methods. Similarly, in the data set of medium-short-term forecasts, the RMSE improvement obtained by $IVMD_{HFCM}$ is also above 70%. For the short-term forecast datasets with relatively small fluctuations, the RMSE improvement is almost all above 60%. Only in the SC dataset, the RMSE of $IVMD_{HFCM}$ increases by 49.32% and 59.49% compared

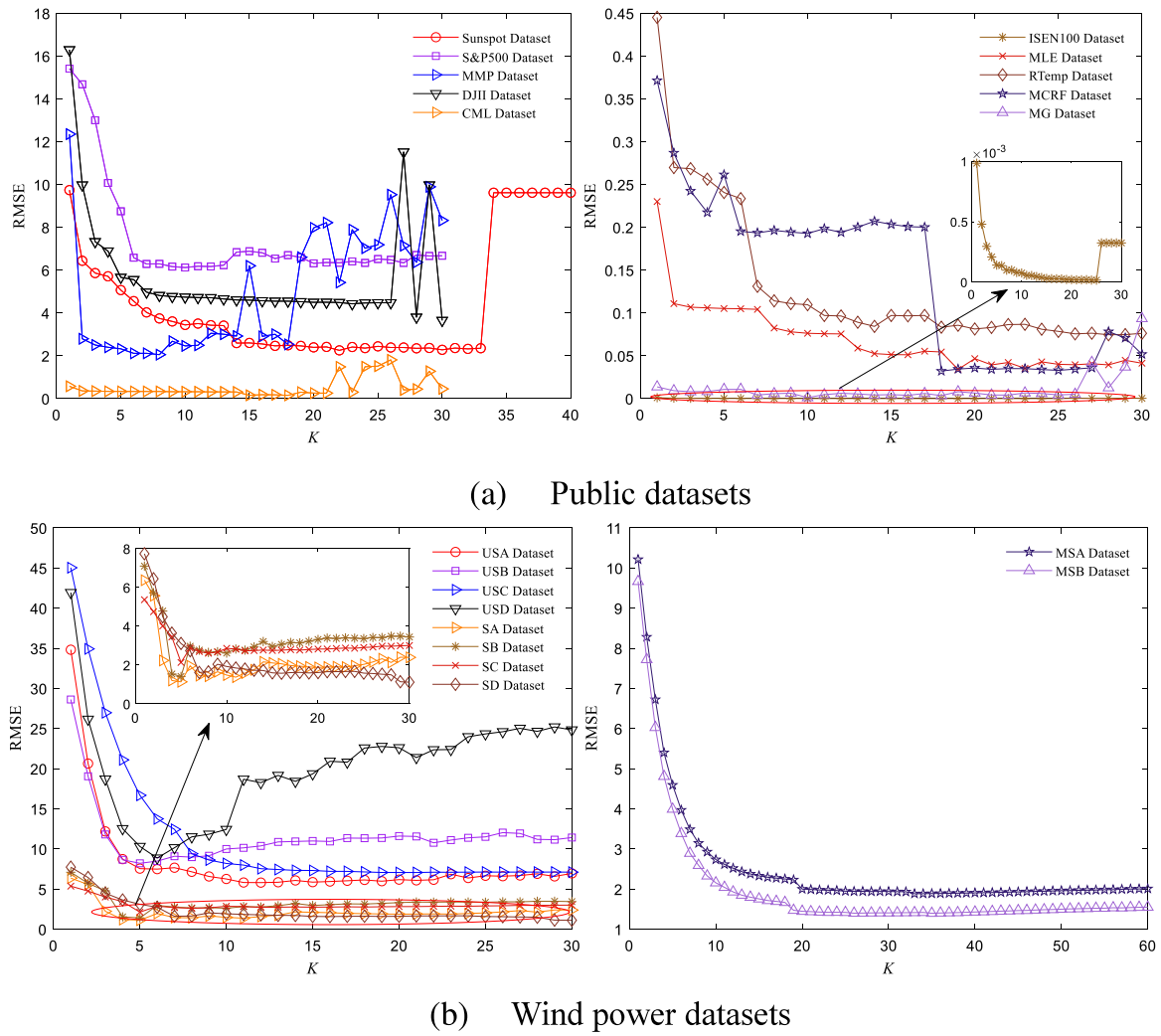
Fig. 6. Variation of RMSE obtained by IVMD_{HFCM} with K .

Table 7

The RMSE and computing time (s) of different algorithms on wind power forecasting.

Datasets	IVMD _{HFCM}	SAE-FCM	EMD-HFCM	Kernel-HFCM	Wavelet-HFCM
USA	6.037 (2298(94))	50.290(4358)	29.431(12)	58.462(2071)	50.858(34)
USB	8.485 (653(15))	40.963(4251)	28.509(13)	42.479(1915)	41.530(29)
USC	7.141 (5040(104))	53.676(7986)	32.804(9)	58.642(1926)	55.269(35)
USD	8.866 (1843(13))	58.179(10792)	36.856(8)	60.543(1870)	61.061(25)
SA	1.132 (2958(18))	5.818(25532)	3.878(33)	6.062(20151)	8.380(157)
SB	1.411 (1258(19))	7.106(6762)	4.476(35)	7.339(15031)	8.421(139)
SC	2.131 (1634(16))	5.358(43373)	4.205(34)	5.261(14501)	6.276(129)
SD	1.715 (10657(110))	6.137(42145)	4.446(28)	6.447(15141)	10.159(143)
MSA	2.000 (32145(203))	10.654(36948)	6.968(26)	10.727(9235)	11.090(90)
MSB	1.447 (31868(213))	10.390(31935)	9.718(26)	10.831(8920)	10.701(91)

Table 8

The improving percentage $P_{RMSE}(\%)$ of IVMD_{HFCM} on wind power forecasting.

Datasets	SAE-FCM	EMD-HFCM	Kernel-HFCM	Wavelet-HFCM
USA	88.00	79.49	89.67	88.13
USB	79.29	70.24	80.03	79.57
USC	86.70	78.23	87.82	87.08
USD	84.76	75.94	85.36	85.48
SA	80.54	70.81	81.33	86.49
SB	80.14	68.48	80.77	83.24
SC	60.23	49.32	59.49	66.05
SD	72.05	61.43	73.40	83.12
MSA	81.23	71.30	81.36	81.97
MSB	86.07	85.11	86.64	86.48

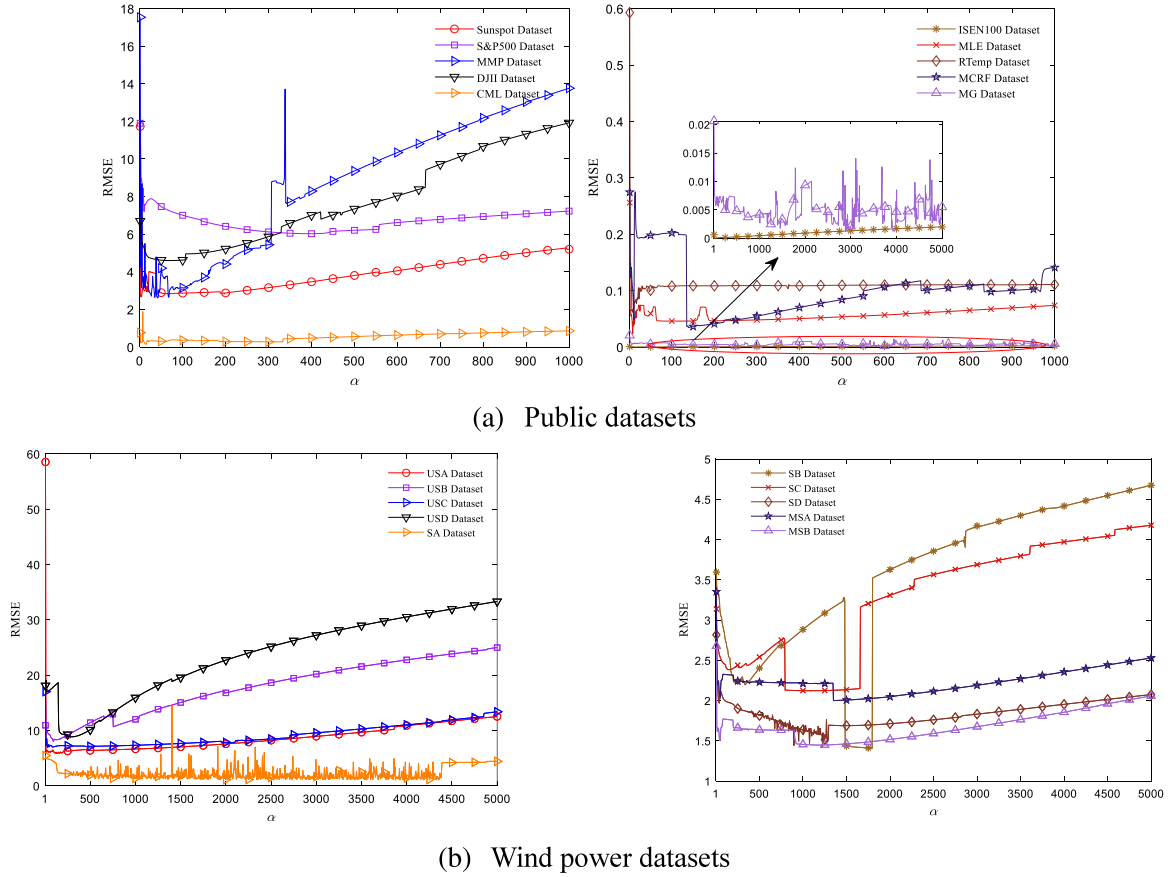
Fig. 7. Variation of RMSE obtained by IVMD_{HFCM} with α .

Table 9

Hyperparameters in wind power forecasting.

Datasets	IVMD _{HFCM} (K, α, h)	SAE-FCM (K, h)	EMD-HFCM (K, h)	Kernel-HFCM (K, h)	Wavelet-HFCM (K, h)
USA	(17, 140, 11)	(5, 2)	(7, 10)	(15, 2)	(3, 2)
USB	(6, 162.5, 4)	(15, 7)	(7, 7)	(2, 2)	(4, 2)
USC	(18, 454, 20)	(40, 7)	(6, 4)	(2, 3)	(4, 3)
USD	(6, 254.5, 8)	(20, 5)	(6, 4)	(2, 4)	(2, 12)
SA	(5, 4150, 8)	(10, 3)	(10, 20)	(2, 17)	(4, 12)
SB	(5, 1754, 16)	(45, 8)	(10, 4)	(2, 19)	(4, 3)
SC	(5, 1416.5, 15)	(40, 7)	(10, 2)	(3, 3)	(4, 12)
SD	(14, 1162.5, 19)	(25, 7)	(9, 5)	(4, 3)	(3, 2)
MSA	(20, 1344, 20)	(20, 19)	(9, 4)	(4, 7)	(4, 2)
MSB	(20, 1186.5, 20)	(15, 13)	(9, 2)	(3, 3)	(4, 2)

with EMD-HFCM and Kernel-HFCM. Even so, the improvement of the IVMD_{HFCM} is relatively significant. Therefore, IVMD_{HFCM} can improve the forecasting accuracy of wind power with high volatility, helping the dispatching agency to accurately make dispatching plans and reduce the economic costs caused by wind power forecasting errors.

The hyperparameters of IVMD_{HFCM} and the order (h) and the number of nodes (K) of HFCM in the contrast methods are shown in Table 9. The actual data of wind power and predicted values obtained by IVMD_{HFCM} are illustrated in Fig. 5, and other methods are shown in Appendix. From Figs. A.1–A.3 we can see, all methods can perceive the changing trend of the time series. However, IVMD_{HFCM} is more able to perceive the data points of the sudden change in the time series, thus it can obtain higher prediction accuracy when forecasting the time series. At the same time, it also shows that IVMD based on sequence frequency decomposition technology can extract more time series features. The evaluation indicators of IVMD_{HFCM} are listed in Table 10. From

Table 10 we can see, the proposed IVMD_{HFCM} is the best compared with other methods in terms of all indicators.

4.5. Hyperparameter analysis

In this section, the sensitive analysis of K , α , and h in IVMD_{HFCM} is conducted. We fix two of the parameters and observe the change in the RMSE obtained by IVMD_{HFCM} when the other parameter changes, and analyze their respective effects on the performance of IVMD_{HFCM}. Besides, the values of the fixed parameter are selected from the results in Tables 5 and 9.

(1) *Effect of K on IVMD_{HFCM}*: The influence of the number K of subseries obtained by IVMD on the performance of IVMD_{HFCM} is analyzed. Here α and h are fixed. RMSE varies with K on all datasets as shown in Fig. 6. For the VMD, too few modes lead to the under segmentation of the data. In Fig. 6, it can be seen that RMSE is the highest in almost all datasets when $K = 1$. Then, as K increases, RMSE reaches a minimum value, and then

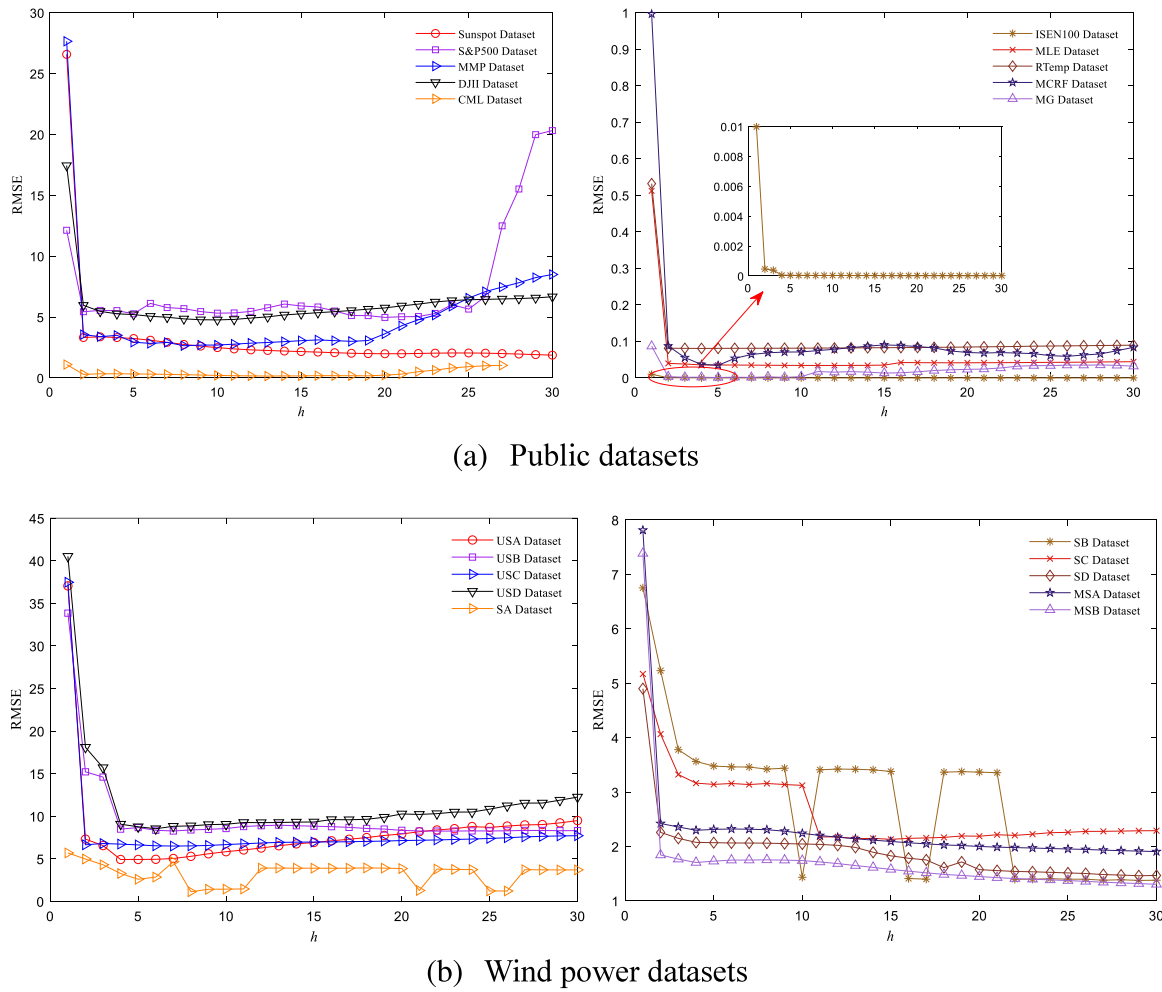
Fig. 8. Variation of RMSE obtained by IVMD_{HFCM} with h .

Table 10

The evaluation indicators of the IVMD_{HFCM} on the wind power datasets.

Datasets	Worst(K, α, h)	Median(K, α, h)	Best(K, α, h)	Average	Std
USA	6.442(17, 241, 11)	6.245(17, 214.5, 11)	6.037(17, 140, 11)	6.245	0.143
USB	8.599(7, 258, 5)	8.545(7, 170.5, 5)	8.485(6, 162.5, 4)	8.555	0.036
USC	7.158(20, 356.5, 20)	7.150(20, 486, 20)	7.141(18, 454, 20)	7.152	0.006
USD	11.208(16, 1397.5, 11)	9.152(6, 262.5, 11)	8.866(6, 254.5, 8)	9.387	0.643
SA	1.415(5, 3706, 8)	1.255(5, 4087, 8)	1.132(5, 4150, 8)	1.277	0.081
SB	1.491(4, 399, 17)	1.448(6, 455, 18)	1.411(5, 1754, 16)	1.448	0.023
SC	2.148(6, 1600, 14)	2.145(6, 1503, 14)	2.131(5, 1416.5, 15)	2.144	0.004
SD	2.289(11, 3351, 8)	1.992(15, 1114.5, 19)	1.715(14, 1162.5, 19)	1.943	0.165
MSA	2.048(20, 1343.5, 20)	2.000(20, 1351.5, 20)	2.000(20, 1344, 20)	2.004	0.014
MSB	1.456(20, 981, 20)	1.456(20, 982, 20)	1.447(20, 1186.5, 20)	1.452	0.004

Table 11

 K and the lowest RMSE obtained by IVMD_{HFCM}.

Datasets	Sunspot	S&P500	MMP	DJII	CML	ISEN100	MLE	RTemp	MCRF	MG
($K, RMSE$)	(22, 2.253)	(10, 6.122)	(8, 2.052)	(30, 3.652)	(18, 0.151)	(25, 1.787)	(19, 0.034)	(28, 0.075)	(18, 0.032)	(10, 0.001)
Datasets	USA	USB	USC	USD	SA	SB	SC	SD	MSA	MSB
($K, RMSE$)	(12, 5.807)	(5, 8.214)	(21, 7.065)	(6, 8.866)	(5, 1.1324)	(5, 1.411)	(5, 2.1311)	(30, 1.115)	(33, 1.885)	(35, 1.397)

RMSE rises if K continues to grow. Besides, in CML, MLE, and MCRF datasets, the phenomenon that the RMSE decreases and then rises as K continues to increase because the decomposition of IVMD is also related to the selection of α . This indicates that if the decomposition number K of the time series continues to increase after increasing to a certain value, the original data and the predicted data will be over-fitted, thus making the HFCM

weight matrix learning inaccurate, and thus increasing the RMSE of the predicted results. Table 11 shows the K value at the lowest RMSE of the test set.

(2) Effect of α on IVMD_{HFCM}: α is the quadratic penalty factor in VMD, which directly affects the accuracy of VMD decomposition. When α changes, the impact of the IVMD method on the prediction accuracy is studied in this subsection. Similarly, K and

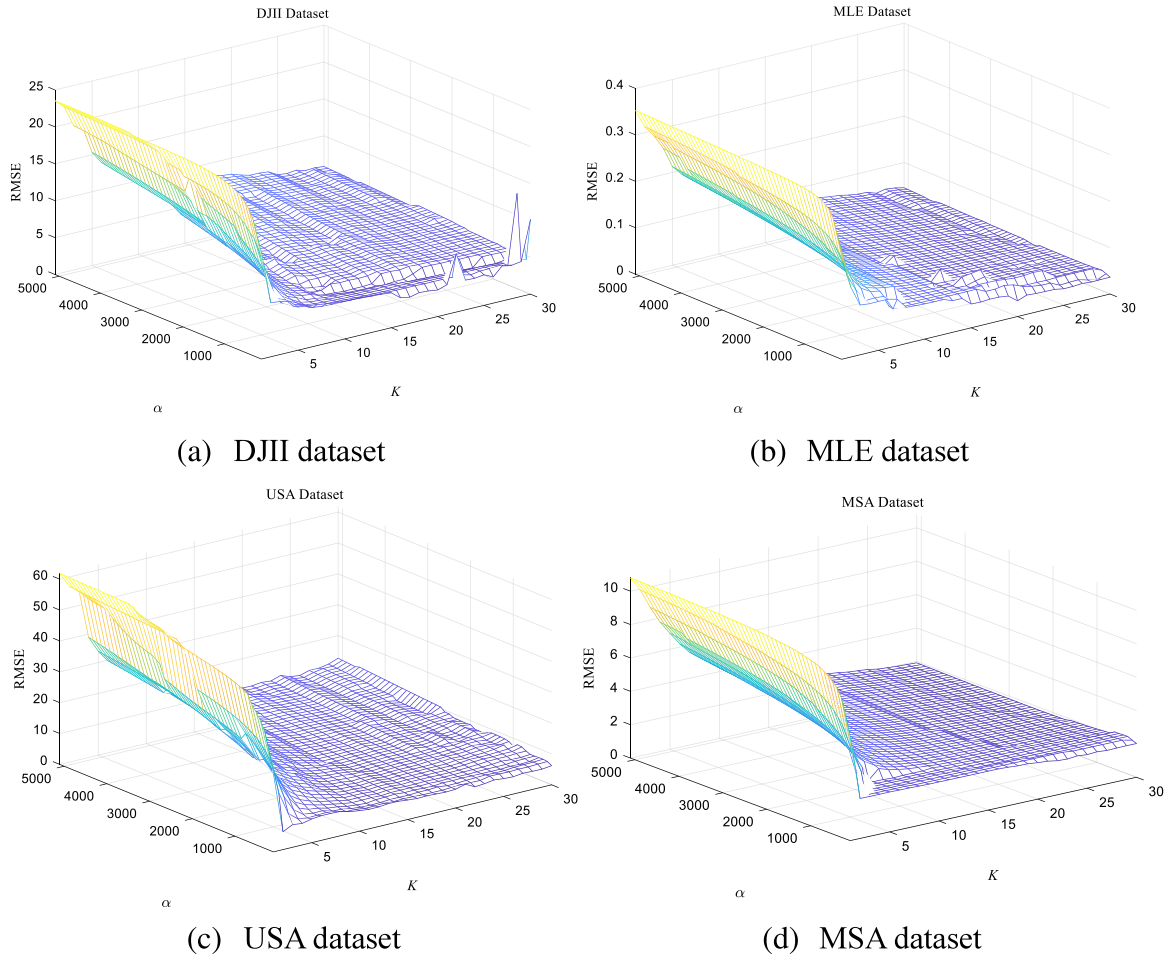


Fig. 9. RMSE versus varying K and α of $IVMD_{HFCM}$ in some datasets.

Table 12

The lowest RMSE obtained by $IVMD_{HFCM}$ and corresponding α .

Datasets	Sunspot	S&P500	MMP	DJII	CML	ISEN100	MLE	RTemp	MCRF	MG
(α , RMSE)	(5, 2.659)	(1975, 6.027)	(200, 2.611)	(345, 4.599)	(65, 0.144)	(5, 2.73e−05)	(35, 0.024)	(35, 0.062)	(690, 0.034)	(2880, 0.001)
Datasets	USA	USB	USC	USD	SA	SB	SC	SD	MSA	MSB
(α , RMSE)	(130, 5.848)	(90, 8.053)	(25, 6.929)	(295, 8.861)	(3870, 1.059)	(1790, 1.405)	(1055, 2.123)	(1220, 1.433)	(1345, 2.000)	(1170, 1.447)

h are fixed. For simplicity, α takes a value every 5 in [1, 5000] to form the hyperparameter pair required in $IVMD_{HFCM}$ with the previously obtained values of K and h . The variation trend of RMSE with α in all datasets is shown in Fig. 7. It can observe that RMSE changes significantly with α . In the SA dataset, the RMSE fluctuates repeatedly with the change of α , which is caused by the large fluctuations of this actual wind power. An appropriate α can make the $IVMD_{HFCM}$ prediction accuracy more accurate. However, with α continually increasing, there will be a knee point when RMSE drops to the lowest value. Therefore, to reduce the prediction error, an appropriate α is necessary in $IVMD_{HFCM}$. The lowest RMSE in all datasets and corresponding α are listed in Table 12.

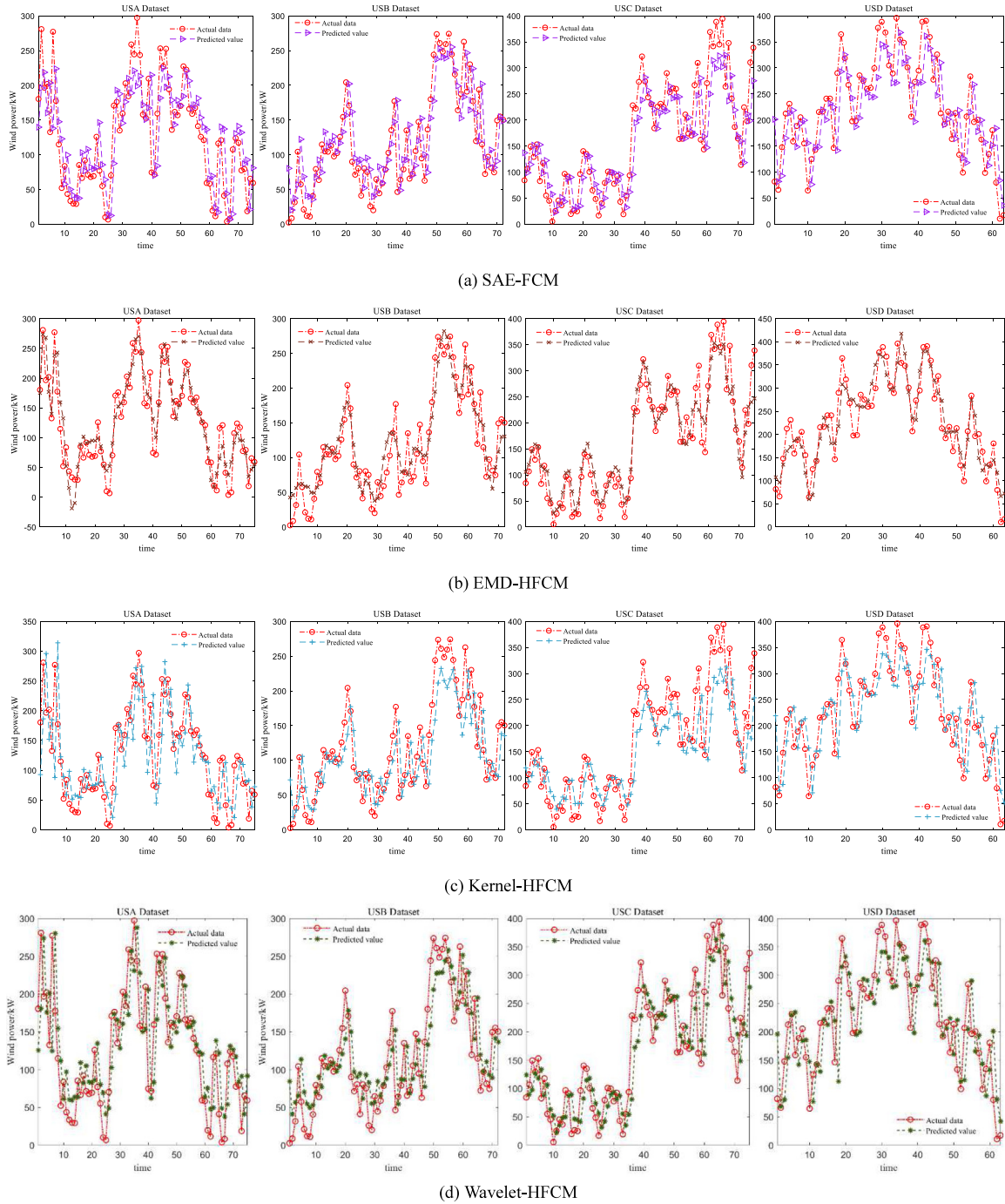
(3) *Effect of h on $IVMD_{HFCM}$* : The effect of order h on the performance of $IVMD_{HFCM}$ is analyzed in all datasets. Fig. 8 shows the trend of RMSE changes with h . It can be noted that when $h = 1$, i.e. the normal FCM, the RMSE is the highest in public datasets and wind power datasets. This indicates that HFCM can indeed improve the prediction performance by using historical

data information. Besides, RMSE will be still decrease as the order h increases, and then fluctuate around a small value. However, when the order h increases to a large value, the predictive performance of $IVMD_{HFCM}$ decreases instead. This is because large order h may get some redundant information from historical data, which leads to over-fitting of the prediction model, thus increasing the error of the prediction results. Therefore, the order h in the $IVMD_{HFCM}$ is not suitable to select a larger value. In addition, as can be seen from Fig. 8, when h is around 15, the RMSE fluctuates around a very small value in all datasets.

(4) *Effect of K and α on $IVMD_{HFCM}$* : K and α are important parameters in VMD, and their effects on $IVMD_{HFCM}$ in DJII, MLE, USA, and MSA datasets are analyzed in this subsection. The value of h is the optimal value found by DE. For simplicity, α takes a value every 100 in [100, 5000], and K takes a value every 1 in [1, 30]. The grid search method is used to record the value of RMSE as it changing with K and α . Fig. 9 shows the variation of RMSE with K and α in four time series. From Fig. 9, when the smaller value of K is selected, no matter what the value of α is, $IVMD_{HFCM}$

Table 13The lowest RMSE obtained by IVMD_{HFCM} and the corresponding K and α .

Datasets	DJII	MLE	USA	MSA
(K, α)	(17, 100)	(29, 200)	(22,100)	(30,2400)
RMSE	4.384	0.310	5.364	1.826

**Fig. A.1.** Forecast results of ultra-short-term wind power datasets by different methods.

will get a larger RMSE. After K and α reach certain values, RMSE only changes around a small value. Therefore, a small K value has a significant effect on IVMD_{HFCM}, while a large value α of has an opposite effect on IVMD_{HFCM}. The K and α corresponding to the minimum RMSE are shown in Table 13.

5. Conclusions

In this paper, a novel forecasting method based on improved VMD and HFCM, namely IVMD_{HFCM}, is proposed to forecast wind power. VMD decomposes the original time series according to its



Fig. A.2. Forecast results of short-term wind power forecasting datasets by different methods.

time–frequency characteristics, so it can effectively extract the feature information of the time series. To accurately decompose the raw time series, the VMD is improved. The number of sub-series obtained by IVMD is used as the nodes of HFCM. Then, the subseries are modeled and forecasted by HFCM, and the Bayesian ridge regression method is adopted to learn the weight of HFCM. The DE algorithm is used to find the optimal hyperparameters of IVMD_{HFCM}, and the indicators show that the RMSE values corresponding to the hyperparameters obtained each time are almost better than those obtained by the comparison methods. Besides, the performance of IVMD_{HFCM} is demonstrated on ten public datasets. The results demonstrate that the performance of IVMD_{HFCM} is superior compared with state-of-art methods. In

wind power datasets, compared with the existing HFCM-based method, the forecasting value obtained by IVMD_{HFCM} is closer to the actual value. Besides, IVMD_{HFCM} can also track the fluctuations of wind power. Consequently, IVMD_{HFCM} can achieve an accurate prediction of wind power and reduce forecast error. The proposed method involves three hyperparameters, and the DE algorithm based on a random population is used to find the optimal parameter pair, but the results obtained are time-consuming. In our future work, the parameter adaptive method will be explored to solve this problem. Moreover, the proposed method only considers the prediction of university time series. However, wind power is affected by a variety of factors. We will also study the application of IVMD_{HFCM} to the forecast of multivariate wind power time series.

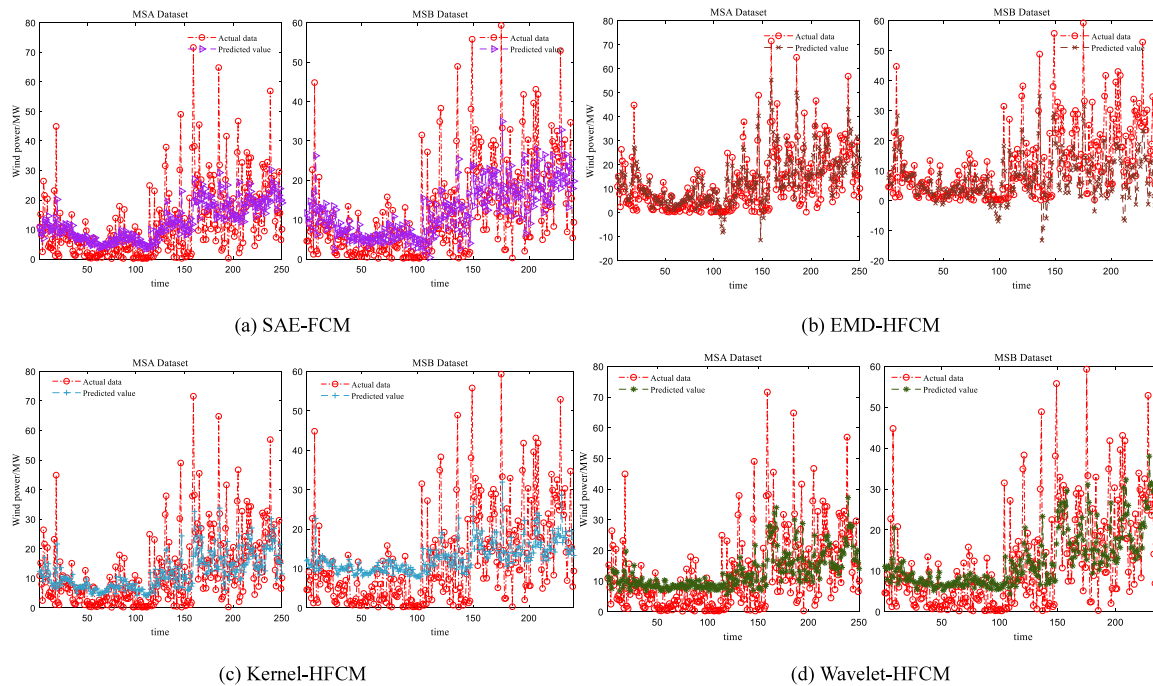


Fig. A.3. Forecast results of medium-short-term forecasting datasets by different methods.

CRediT authorship contribution statement

Baihao Qiao: Performed the experiments, Data curation, Investigation, Writing – original draft. **Jing Liu:** Designed the experiments, Methodology, Writing – review & editing. **Peng Wu:** Investigation, Data curation. **Yingzhi Teng:** Data curation, Method.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

Appendix presents the forecasting results of the comparative methods including SAE-FCM, EMD-HFCM, Kernel-HFCM, and Wavelet-HFCM used on different wind power datasets. Figs. A.1–A.3 shows the forecasting results of the comparison method on the ultra-short-term wind power datasets, short-term wind power datasets, and medium-short-term wind power datasets, respectively.

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